

Project Initialization and Planning Phase

Date	6 July 2026
Team ID	
Project Title	Plugging into the Future : An Exploration of Electricity consumption Patterns
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

Project Overview	
Objective	To develop a Machine Learning-based Electricity Consumption Forecasting System that predicts future power utilization patterns accurately and efficiently to help in smart grid management.
Scope	The project analyzes historical power usage datasets, preprocesses time-series data, trains forecasting models, and provides real-time electricity consumption predictions and analytics through a web application.
Problem Statement	
Description	Traditional grid monitoring lacks predictive foresight, leading to high energy wastage, unpredictable load spikes, and inefficient power distribution due to the lack of automated consumption pattern analytics.
Impact	The proposed system improves grid reliability, reduces energy wastage, increases prediction accuracy for future demands, and enhances overall consumer energy-saving efficiency.

Proposed Solution	
Approach	The system uses Machine Learning regression algorithms to analyze time-series smart meter records and household energy metrics to predict future power consumption trends and peak demand loads.
KeyFeatures	Machine Learning-based electricity load forecasting Time-series data preprocessing and feature engineering Real-time demand visualization dashboard using Flask

Resource Requirements

ResourceType	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU	Intel i5 or above
Memory	RAM specifications	8GB
Storage	Disk space for data,models,and logs	256GBSSD or above
Software		
Frameworks	Python frame works	Flask
Libraries	M L & Data Analysis	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, VS code / PYCharm
Data		
Data	Source,size,format	Household Electricity Consumption Dataset (CSV), UCI Machine Learning Repository / Kaggle